

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (CL) Examination

November 2013

CL402 Computer Aided Structural Analysis

Date: 16.11.2013, Saturday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) Analyse the given portal frame using flexibility method member approach. Derive B_{MS} and F_M matrix. Refer Fig. 1. [07]

Q - 2 (a) Analyse the given 2-span continuous beam using direct stiffness method. Refer Fig. 2. [14]

OR

Q - 2 (a) Analyse the given 2-span continuous beam using direct stiffness method. Refer Fig. 3. [14]

Q - 3 (a) Analyse the given portal frame using direct stiffness method. Refer Fig. 4. [14]

OR

Q - 3 (a) Develop stiffness matrix (3 x 3) size for space frame assembly by direct method with usual notation. Refer Fig. 5 [04]

(b) (i) Explain symmetry, anti-symmetry and un-symmetry using examples. [10]

(ii) Explain Rotation of Axes in Two Dimensions.

SECTION - II

Q - 4 (a) Analyse the following beams using flexibility method. Refer Fig. 6. [07]

Q - 5 (a) Assemble Jacobian matrix and strain displacement matrix corresponding to the Gauss point (0.6740, 0.6740) for the element shown in Fig: 7. [09]

(b) Determine the Cartesian co-ordinate of the point P ($\xi = 0.45$, $\eta = 0.65$) shown in Fig:8 [05]

OR

Q - 5 (a) Derive shape functions for nine noded rectangular element having width = 2 and depth = 2. Use natural co-ordinate system. [09]

(b) For the iso-parametric quadrilateral element shown in Fig: 9 determine local co-ordinates of the point Q which has Cartesian co-ordinate (7, 4). [05]

Q - 6 (a) Analyse the given 2-span continuous beam using flexibility approach. Refer Fig. 10 [14]

OR

Q - 6 (a) Analyse the given rectangular pin-jointed structure using flexibility approach. Refer Fig. 11 [14]

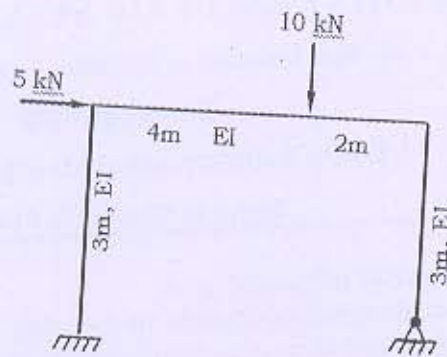


Fig. 1

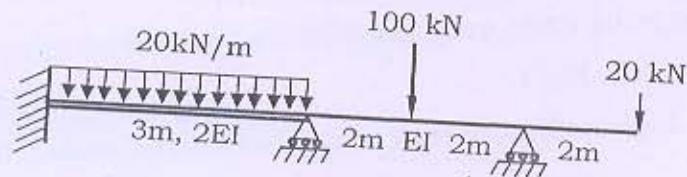


Fig. 2

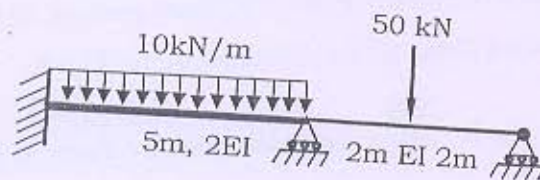


Fig. 3

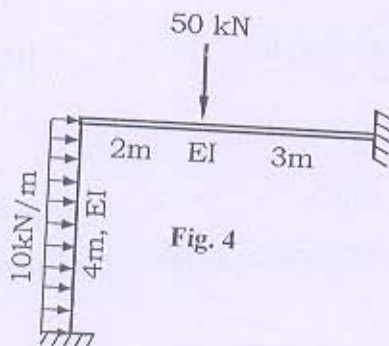


Fig. 4

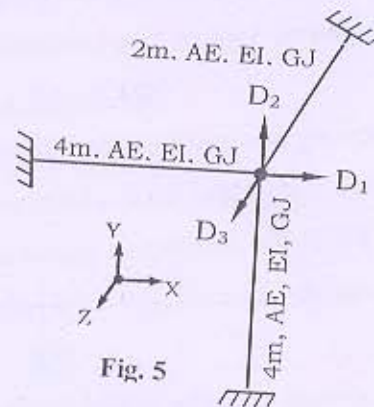


Fig. 5

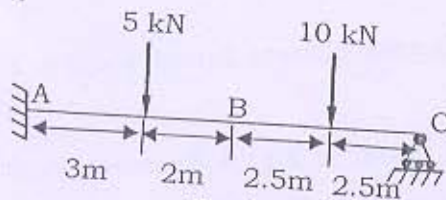


Fig. 6

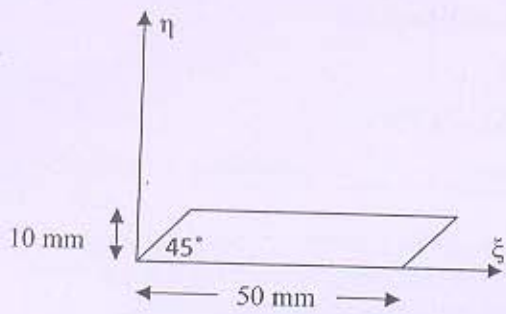


Fig. 7

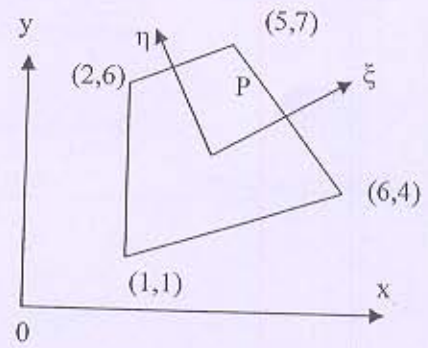


Fig. 8

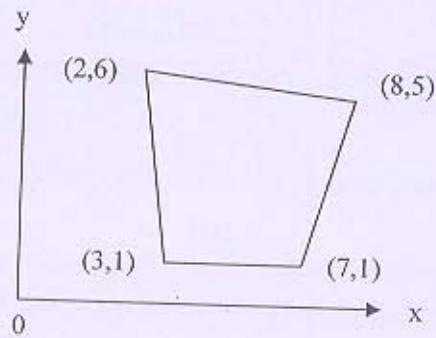


Fig. 9

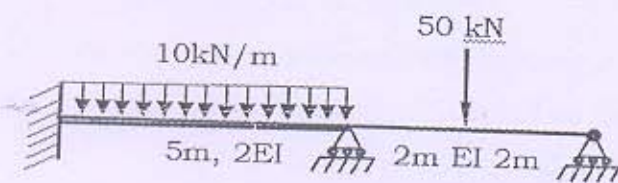


Fig. 10

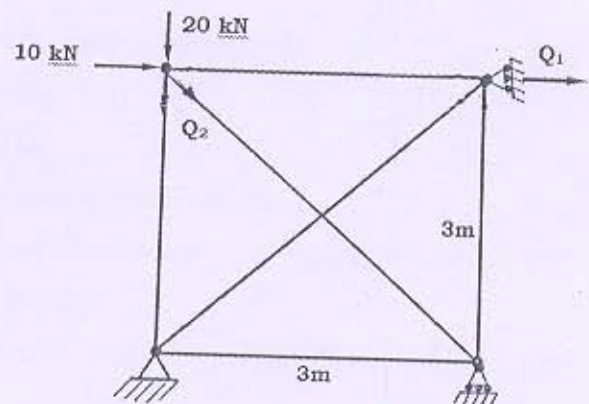


Fig.11
